

Data Visualization Assignment

Kathryn Gour

Getting Started

This assignment will test your ability to use the tidyverse to perform a complete data analysis workflow: from data wrangling to publication-quality visualization. You will be working with datasets from the psych package, which is commonly used in psychological research.

Your goal for each problem is to write a single, clean dplyr pipe (%>%) that creates the requested data, and then pipe that data directly into a ggplot() call to create a publication-quality plot.

This assignment uses datasets from the **psych** package. For each problem: - Briefly **describe your wrangling** (1–2 sentences). - Complete the r code in the # TODO sections. - Use tidy verbs (**filter**, **mutate**, **select**, **summarise**, **pivot_***). - Choose **appropriate plots** (when asked) that match variable types and questions. - Be sure to polish chart elements! For example, label numeric levels when necessary (e.g., 1 = Male, 2 = Female) - Be sure to comment your code!

Install/Load Required Packages

```
if (!require(psych)) install.packages("psych")
if (!require(tidyverse)) install.packages("tidyverse")
if (!require(corrplot)) install.packages("corrplot")
if (!require(ggplot2)) install.packages("ggplot2")
if (!require(janitor)) install.packages("janitor")

library(psych)
library(tidyverse)
library(corrplot)
library(ggplot2)
library(janitor) # i like this package bc it makes it sm easier to manip vectors
```

Part 1: Exploring Distributions (bfi Dataset)

The bfi dataset contains 25 personality items from the Big Five Inventory (BFI), along with gender, education, and age.

1) Distribution of Agreeableness

Data: 'psych::bfi'

Goal Create a density plot showing the distribution of the mean “Agreeableness” score.

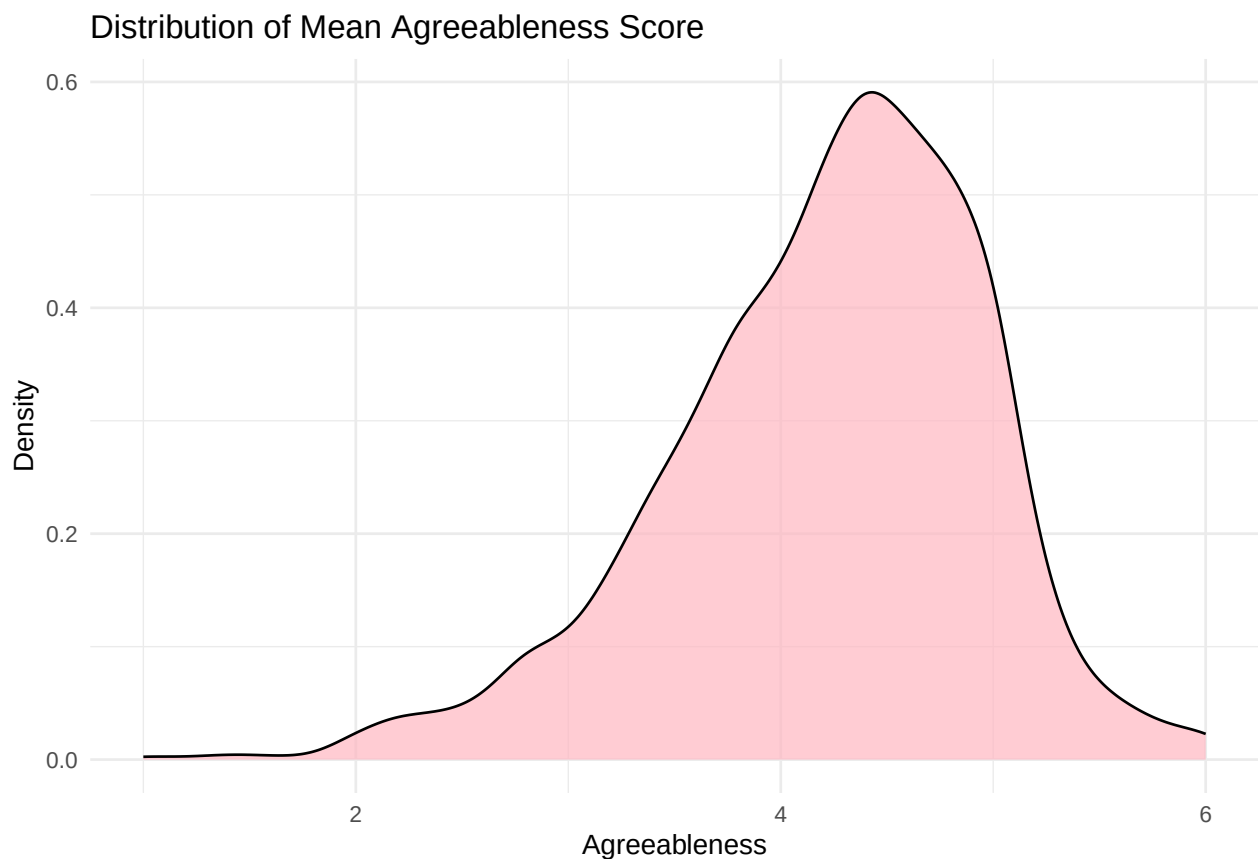
Wrangling: Start with the bfi data. Use mutate() to create a new agreeableness column by calculating the row mean of the five agreeableness items (A1, A2, A3, A4, A5). Make sure to handle missing values (na.rm = TRUE).

Plotting: Pipe the wrangled data into `ggplot()`. * Use `geom_density()` to plot the agreeableness distribution.

- Add a fill to the geom (e.g., “skyblue”) and set `alpha = 0.7`.
- Add appropriate labels using `labs()` (title, subtitle, x-axis).
- Use `theme_minimal()`.

```
#YOUR CODE HERE
bfi_wrangled <- psych::bfi %>%
  # compute mean scores for agreeableness using rowMeans
  mutate(Agreeableness = rowMeans(select(., A1:A5), na.rm=TRUE))

# density plot of mean agreeableness
ggplot(bfi_wrangled, aes(x = Agreeableness)) +
  geom_density(fill = "lightpink", alpha = 0.7) +
  labs(
    title = "Distribution of Mean Agreeableness Score",
    x = "Agreeableness",
    y = "Density"
  ) +
  theme_minimal()
```



Describe the Result: The density plot shows that participants often scored around 4.5/6 on the agreeableness subscale of the BFI. —

2) Neuroticism by Gender

Data: ‘psych::bfi

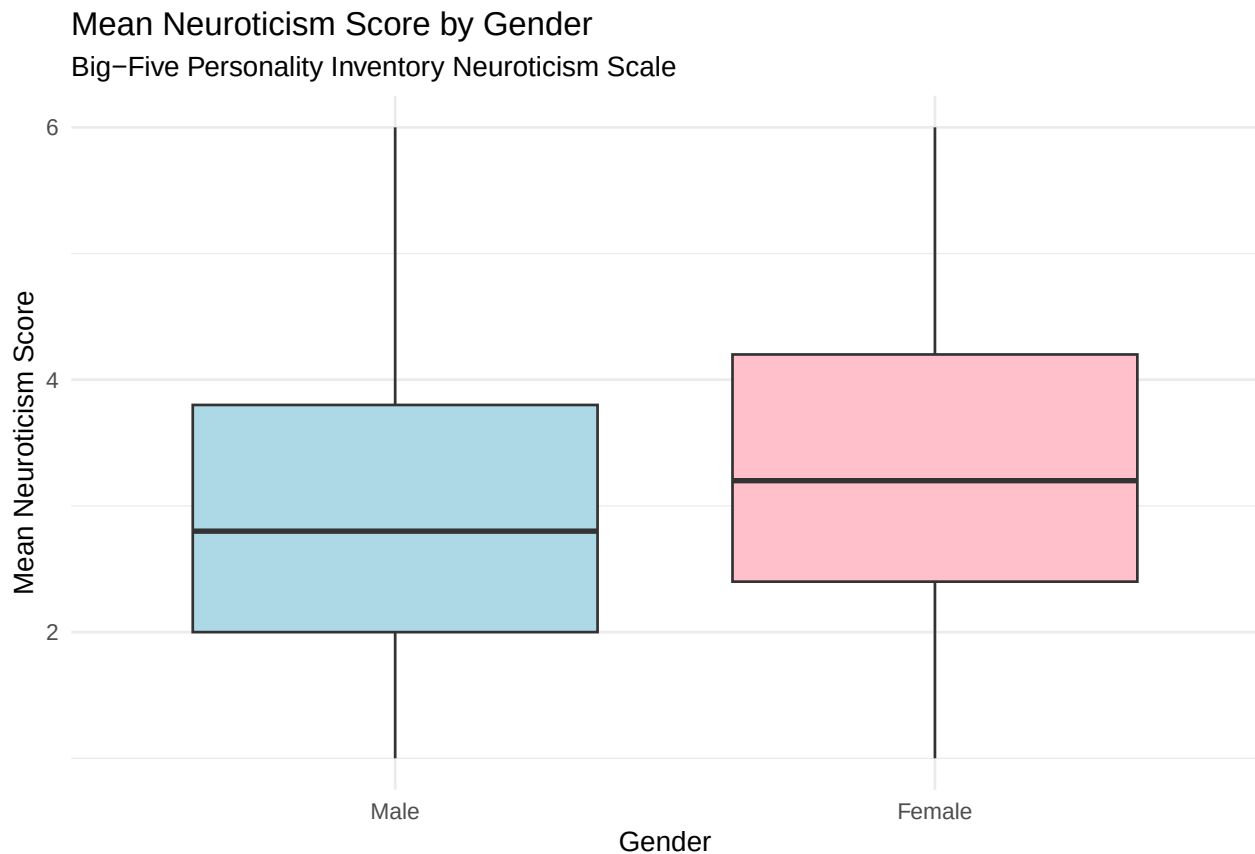
Goal Create a boxplot comparing the distribution of mean “Neuroticism” scores for males and females.

Wrangling: Start with `bfi`. Create a neuroticism column (mean of N1...N5). Then, `filter()` out any NA values for gender. Finally, `mutate()` the gender column into a factor with descriptive labels (e.g., 1 = “Male”, 2 = “Female”).

Plotting: Use `geom_boxplot()` to map gender to the x-axis and neuroticism to the y-axis.

- Add `fill = gender` to the `aes()` mapping.
- Add `labs()` (e.g., title, subtitle, `x`=“Gender”, `y`=“Mean Neuroticism Score”).
- Use `theme_minimal()` and remove the legend (`theme(legend.position = “none”)`).

```
#YOUR CODE HERE
bfi_wrangled2 <- psych::bfi %>%
  filter(!is.na(gender)) %>%
  # compute neuroticism average
  mutate(
    Neuroticism = rowMeans(select(., N1:N5), na.rm = TRUE),
    gender = factor(gender,
                    levels = c(1, 2),
                    labels = c("Male", "Female")) # Add labels for gender
  )
# boxplot with mean neuroticism score by gender
ggplot(data = bfi_wrangled2, aes(x = gender, y = Neuroticism, fill=gender)) +
  geom_boxplot() +
  scale_fill_manual(values=c('lightblue','pink')) +
  labs(title = "Mean Neuroticism Score by Gender",
       subtitle = "Big-Five Personality Inventory Neuroticism Scale",
       x = "Gender", y = "Mean Neuroticism Score") +
  theme_minimal() +
  theme(legend.position = "none")
```



Describe the Result: According to the boxplot, it seems like women have a higher mean neuroticism score compared to men, as measured by the BFI. —

3) Conscientiousness by Education Level

Data: `psych::bfi`

Goal: Create a column chart showing the average “Conscientiousness” score by education level and showing the 95% confidence interval.

Wrangling: Create a conscientiousness column (mean of C1...C5). `filter()` out NA education levels. `group_by()` education. `summarise()` to find the `mean_c_score`.

Plotting: Use `geom_col()` to map education to x and `mean_c_score` to y. (Note: education should be a factor to prevent ggplot from treating it as a continuous number. Use `factor(education)` in your `aes()` mapping).

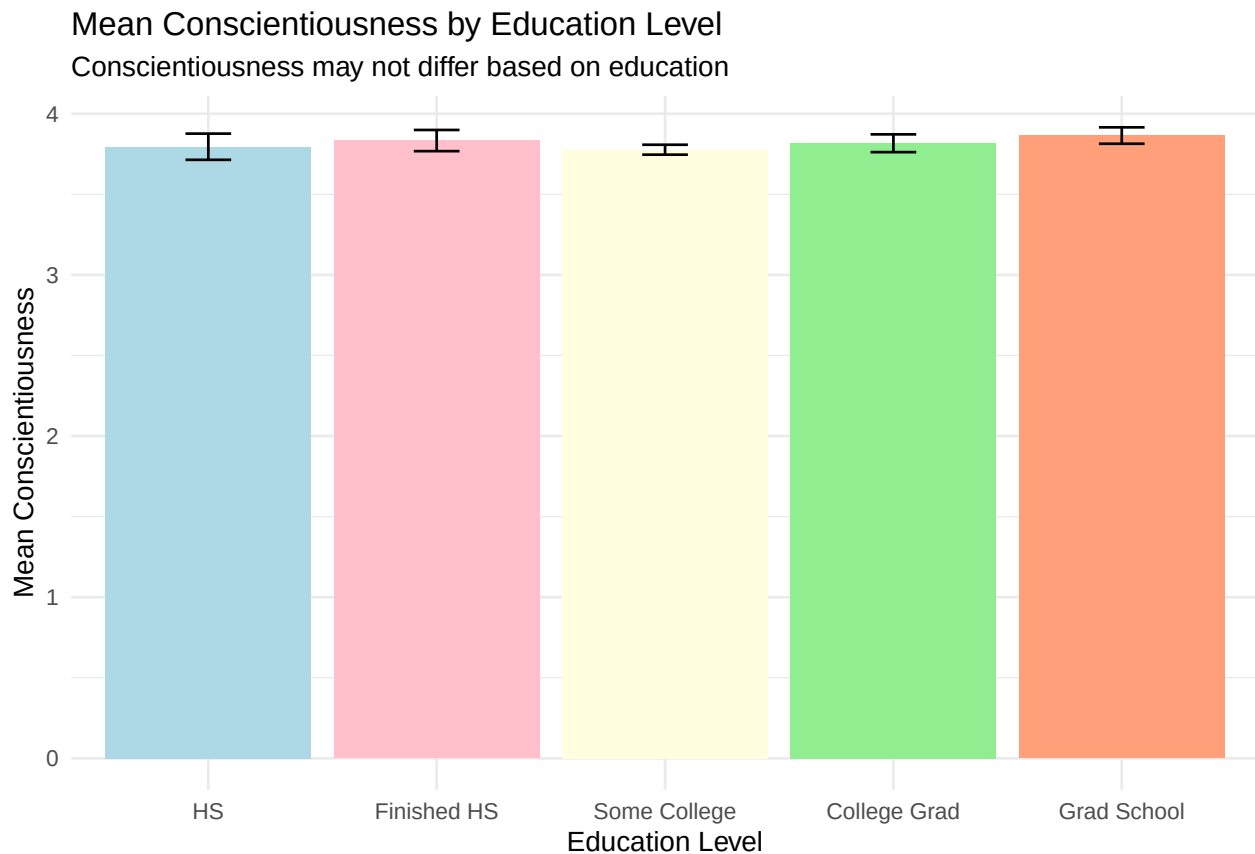
- Add `fill = factor(education)` to `aes()`.
- Add `labs()` (title, subtitle, `x`=“Education Level”, `y`=“Mean Conscientiousness”).
- Use `geom_errorbar()` to display the 95% CI for each mean.
- Use `scale_x_discrete()` to provide meaningful labels for the education levels (1=HS, 2=Finished HS, 3=Some College, 4=College Grad, 5=Grad School).

```
# Example use of scale_x_discrete()
#scale_x_discrete(
#   limits = c("3", "2", "1"), # limits controls order of factors
#   labels = c("Subcompact", "Compact", "Midsize") # create labels
# )
```

#YOUR CODE HERE

```
# we are making so many bfi_wrangleds lol
# we can use group_by() and summarise() here; then use the ci functions!
bfi_wrangled3 <- psych::bfi %>%
  filter(!is.na(education)) %>%
  mutate(
    Conscientiousness = rowMeans(select(., C1:C5), na.rm = TRUE)
  ) %>%
  group_by(education) %>%
  summarise(
    mean_c_score = mean(Conscientiousness, na.rm = TRUE),
    se = sd(Conscientiousness, na.rm = TRUE) / sqrt(n()), # standard error
    ci_lower = mean_c_score - 1.96 * se,
    ci_upper = mean_c_score + 1.96 * se
  )

# i used this r color palette cheatsheet for my fyp but figured if you share
# the link with the class they'd appreciate it
# https://nceas.ucsb.edu/sites/default/files/2020-04/colorPaletteCheatsheet.pdf
ggplot(bfi_wrangled3, aes(x = factor(education), y = mean_c_score,
  fill = factor(education))) +
  scale_fill_manual(values=c('lightblue','pink','lightyellow','lightgreen',
    'lightsalmon')) +
  geom_col() +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), width = 0.2) +
  scale_x_discrete(
    labels = c( "1" = "HS", "2" = "Finished HS", "3" = "Some College",
      "4" = "College Grad", "5" = "Grad School")) +
  labs(
    title = "Mean Conscientiousness by Education Level",
    subtitle = "Conscientiousness may not differ based on education",
    x = "Education Level",
    y = "Mean Conscientiousness"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Describe the Result: The bar plot indicates that mean conscientiousness levels do not seem to vary greatly based on education level. — # Part 2: Visualizing Relationships (sat.act Dataset)

4) SAT Verbal vs. SAT Quantitative

Data: `psych::sat.act`

Goal: Create a scatterplot to visualize the relationship between SAT Verbal (SATV) and SAT Quantitative (SATQ) scores.

Wrangling: None needed, just pipe `sat.act` directly into `ggplot()`.

Plotting: Use `geom_point()` to map SATV to x and SATQ to y. Set `alpha = 0.5` to handle overplotting.

- Add a linear regression line using `geom_smooth()`.
- Add `labs()` (e.g., title, subtitle, x="SAT Verbal", y="SAT Quantitative").
- Use `theme_bw()`.

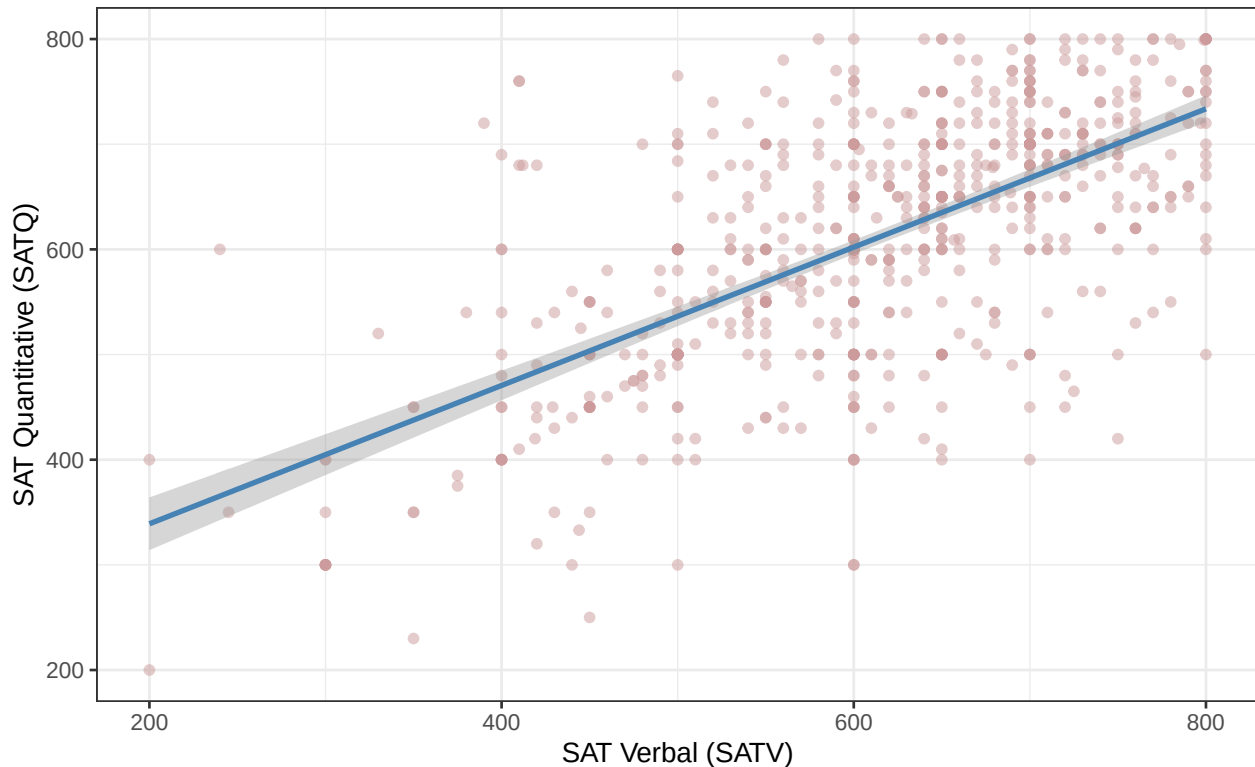
#YOUR CODE HERE

```
ggplot(psych::sat.act, aes(x = SATV, y = SATQ)) +
  geom_point(color = "rosybrown3", alpha = 0.5) +
  geom_smooth(method = "lm", color = "steelblue") +
  labs(
    title = "Relationship Between SAT Verbal and SAT Quantitative Scores",
    subtitle = "SATV and SATQ scores have a positive, linear relationship",
    x = "SAT Verbal (SATV)",
    y = "SAT Quantitative (SATQ)"
  )
```

```
) +  
theme_bw()
```

Relationship Between SAT Verbal and SAT Quantitative Scores

SATV and SATQ scores have a positive, linear relationship



Describe the Result: According to our scatterplot, we can see that there is a clear positive, linear relationship between SAT Verbal performance and SAT Quantitative performance. Our regression line demonstrates this as well. —

5) Faceting the SAT Relationship by Gender

Data: `psych::sat.act`

Goal: Expand on Problem 4. Does the relationship between SATV and SATQ differ by gender?

Wrangling: `mutate()` the gender column to a factor with labels (1="Male", 2="Female").

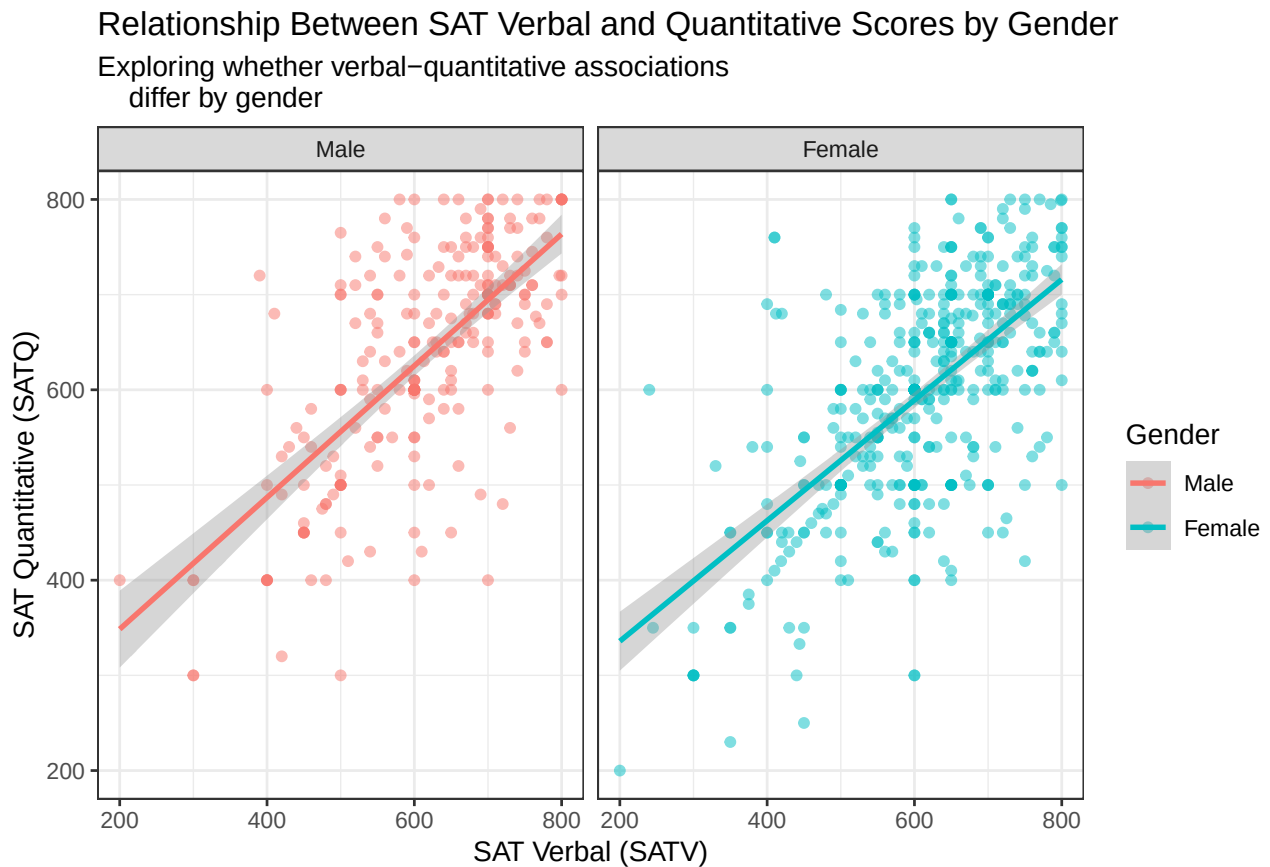
Plotting: Create the same plot as Problem 4 (`geom_point` + `geom_smooth`).

- Add `color = gender` to the `aes()` in the main `ggplot()` call.
- Add `facet_wrap(~ gender)` to create two separate plots.
- Use `labs()` and `theme_bw()`.

#YOUR CODE HERE

```
satact_wrangled <- psych::sat.act %>%  
  # convert gender codes to labeled factors  
  mutate(gender = factor(gender, levels = c(1, 2),  
                          labels = c("Male", "Female")))  
  
ggplot(satact_wrangled, aes(x = SATV, y = SATQ, color = gender)) +  
  geom_point(alpha = 0.5) +
```

```
geom_smooth(method = "lm", se = TRUE) +
facet_wrap(~ gender) +
labs(
  title = "Relationship Between SAT Verbal and Quantitative Scores by Gender",
  subtitle = "Exploring whether verbal-quantitative associations
  differ by gender",
  x = "SAT Verbal (SATV)",
  y = "SAT Quantitative (SATQ)",
  color = "Gender",
) +
theme_bw()
```



Describe the Result: It does not seem like there is an apparent difference in the positive, linear verbal-quantitative association, when comparing between genders. —

6) ACT Scores by Education

Data: `psych::sat.act`

Goal: Create a bar chart showing the average ACT score by education level, ordered from highest to lowest.

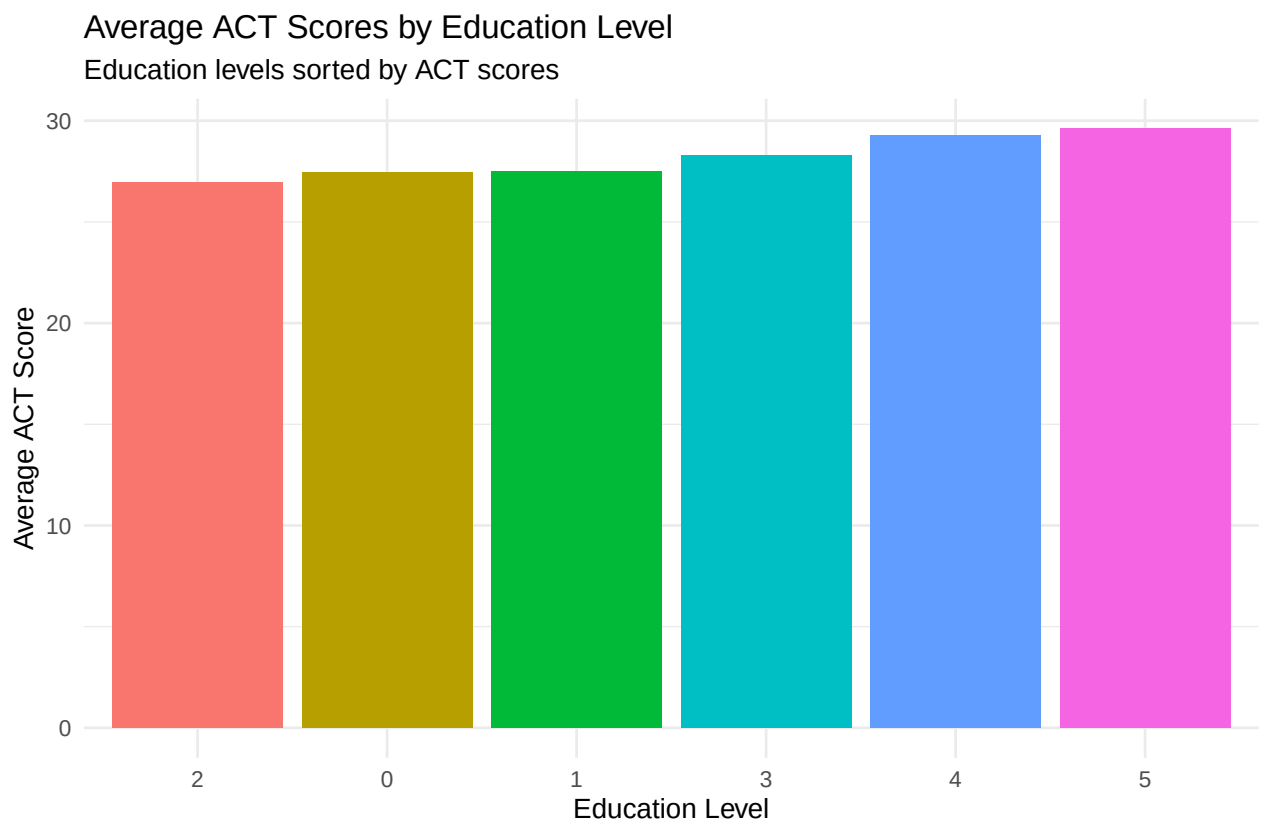
Wrangling: `group_by(education)`, `summarise(mean_act = mean(ACT, na.rm = TRUE))`. Use `mutate()` to `reorder()` the education factor by `mean_act` (e.g., `education = reorder(factor(education), mean_act)`).

Plotting: Use `geom_col()` to map education to x and `mean_act` to y.

- Add `fill = education` to the `aes()` and remove the legend.
- Add `labs()` and `theme_minimal()`.


```
#YOUR CODE HERE
satact_wrangled2 <- psych::sat.act %>%
  group_by(education) %>%
  summarise(mean_act = mean(ACT, na.rm = TRUE)) %>%
  mutate(education = reorder(factor(education), mean_act))

ggplot(satact_wrangled2, aes(x = education, y = mean_act, fill = education)) +
  geom_col() +
  labs(
    title = "Average ACT Scores by Education Level",
    subtitle = "Education levels sorted by ACT scores",
    x = "Education Level",
    y = "Average ACT Score",
    caption = "Data source: psych::sat.act dataset"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Data source: psych::sat.act dataset

Describe the Result: It seems like there is not a great difference in average ACT score sorted by on education level. —

7) BFI: Trait Tradeoffs (E vs N)

Data: psych::bfi

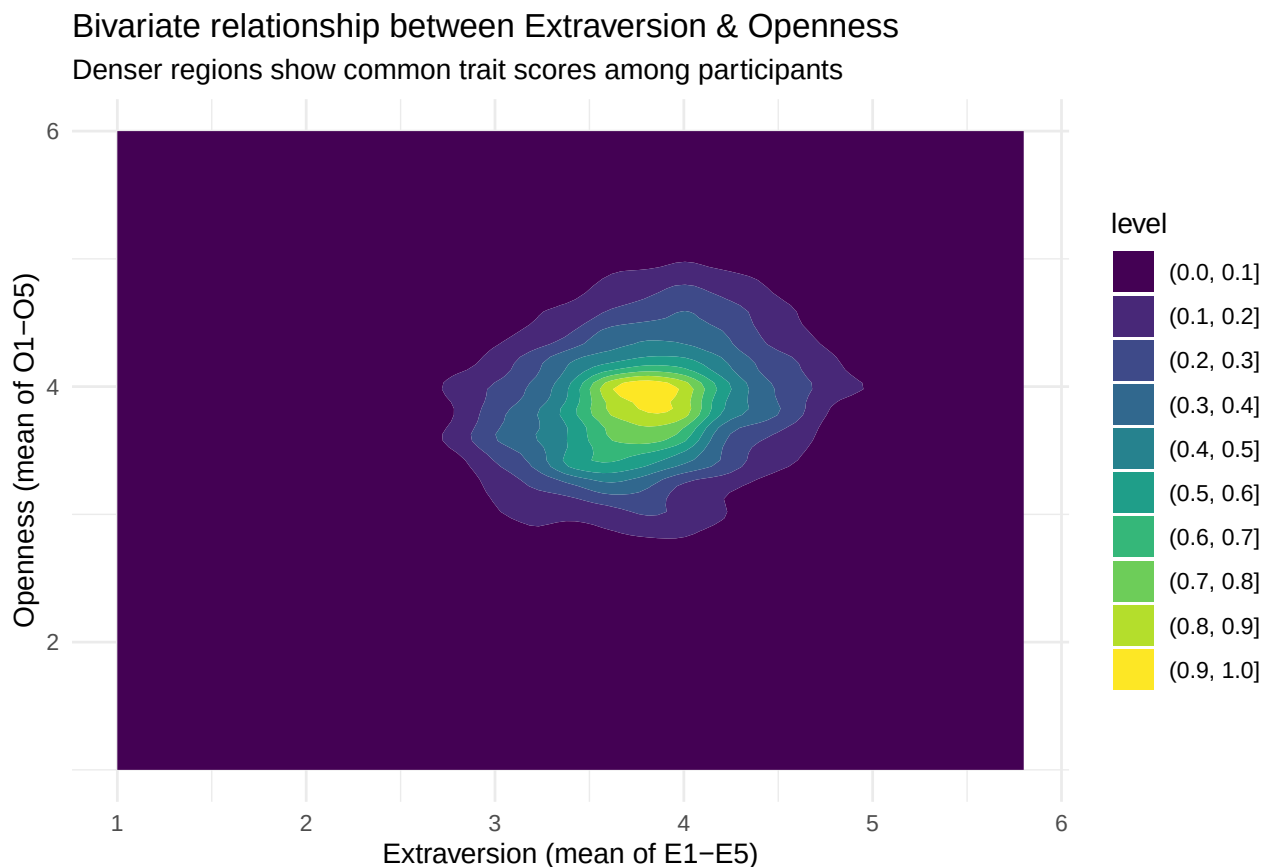
Goal: Compute mean **Extraversion** and **Openness** and visualize their **bivariate relationship**. Pick one: **2D density (filled)**, or **scatter + ggMarginal**. Briefly interpret the pattern.

Wrangling: `summarize()` (e.g., mean of E1...E5).

Plotting: Use `geom_density_2d_filled()` or `ggMarginal()`.

```
#YOUR CODE HERE
# should make a new data pipeline name but i committed to the bit at this point
bfi_wrangled4 <- psych::bfi %>%
  mutate(extraversion = rowMeans(select(., E1:E5), na.rm = TRUE),
         openness = rowMeans(select(., O1:O5), na.rm = TRUE))

ggplot(bfi_wrangled4, aes(x = extraversion, y = openness)) +
  geom_density_2d_filled(contour_var = "ndensity") +
  labs(
    title = "Bivariate relationship between Extraversion & Openness",
    subtitle = "Denser regions show common trait scores among participants",
    x = "Extraversion (mean of E1-E5)",
    y = "Openness (mean of O1-O5)",
  ) +
  theme_minimal()
```



Describe the Result: It seems like the most common, mean trait score for Extraversion and Openness was around (4, 4). This indicates that people who score high on the extraversion subscale of the BFI might also score high on the openness subscale. —

8) Visualizing Intercorrelations Among Cognitive Ability Tests

Data: `ability.cov`

Goal: Convert the `ability.cov` covariance matrix in `$cov` to a correlation matrix using `cov2cor()` and create a

correlogram using `corrplot`.

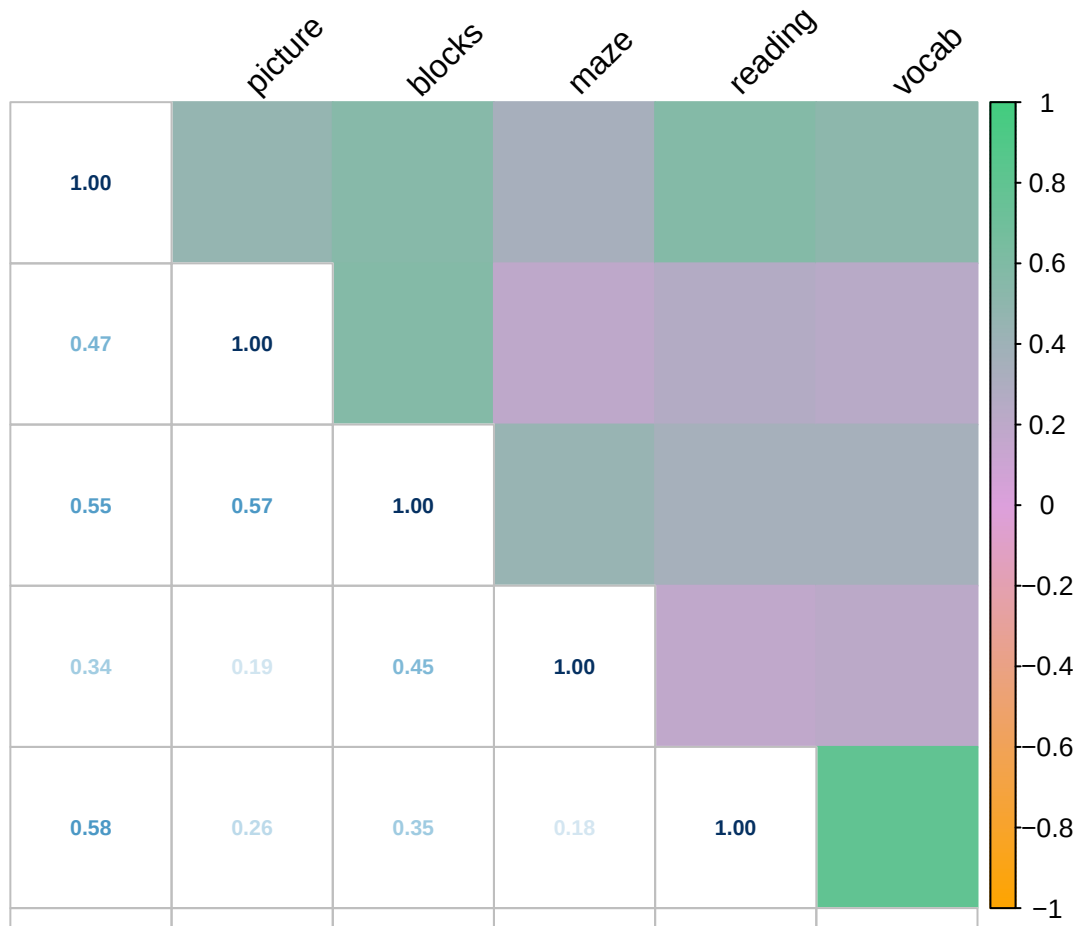
- The top half of the matrix should represent the correlations using shapes or colors and the bottom half of the matrix should contain the values.
- Change the default colors from the default palette

```
#YOUR CODE HERE
# convert covariance matrix from the 'ability.cov' dataset to a corr matrix
ability_cov <- ability.cov$cov
ability_cor <- cov2cor(ability_cov)

# custom colors
corr_colors <- colorRampPalette(c("orange", "plum", "seagreen3"))(200)

# create correlogram: colors (upper), coefficients (lower)
corrplot(ability_cor,
  method = "color",
  type = "upper",
  col = corr_colors,           # custom color palette
  tl.col = "black",           # label color
  tl.srt = 45,                 # rotate text labels
  diag = FALSE                 # no diagonal
)

# add numeric correlations on the lower triangle
corrplot(
  ability_cor,
  add = TRUE,                  # overlay on existing plot
  type = "lower",              # lower half
  method = "number",           # show numeric values
  tl.pos = "n",                # suppress labels this time
  number.cex = 0.7             # adjust size of numbers
)
```



```
# # Compute correlation test results (r and p)
# corr_test <- psych::corr.test(bfi_items) # Use corr.test from psych package
# p_mat <- corr_test$p # extract p-values
#
# # Plot only significant correlations
# corplot(bfi_cor, p.mat = p_mat, sig.level = 0.01, insig = "blank",
#         method = "color",
#         type = "upper",
#         col = colorRampPalette(c("blue", "green", "red"))(200),
#         title = "Only Significant Correlations (p < .01)",
#         mar = c(0,0,2,0))
```

Describe the Result: The correlogram indicates that the strongest, positive association between cognitive ability tests is that of reading and vocab. —

9) SAT & ACT: Relationships

Data: `psych::sat.act` (columns typically SATV, SATQ, ACT)

Goal: Tidy column names, then visualize relationships:

Plotting: Set up the `aes()` to plot SATV vs SATQ with ACT as size.

- Add a smooth trend. Comment on any nonlinearity.
- Add labels

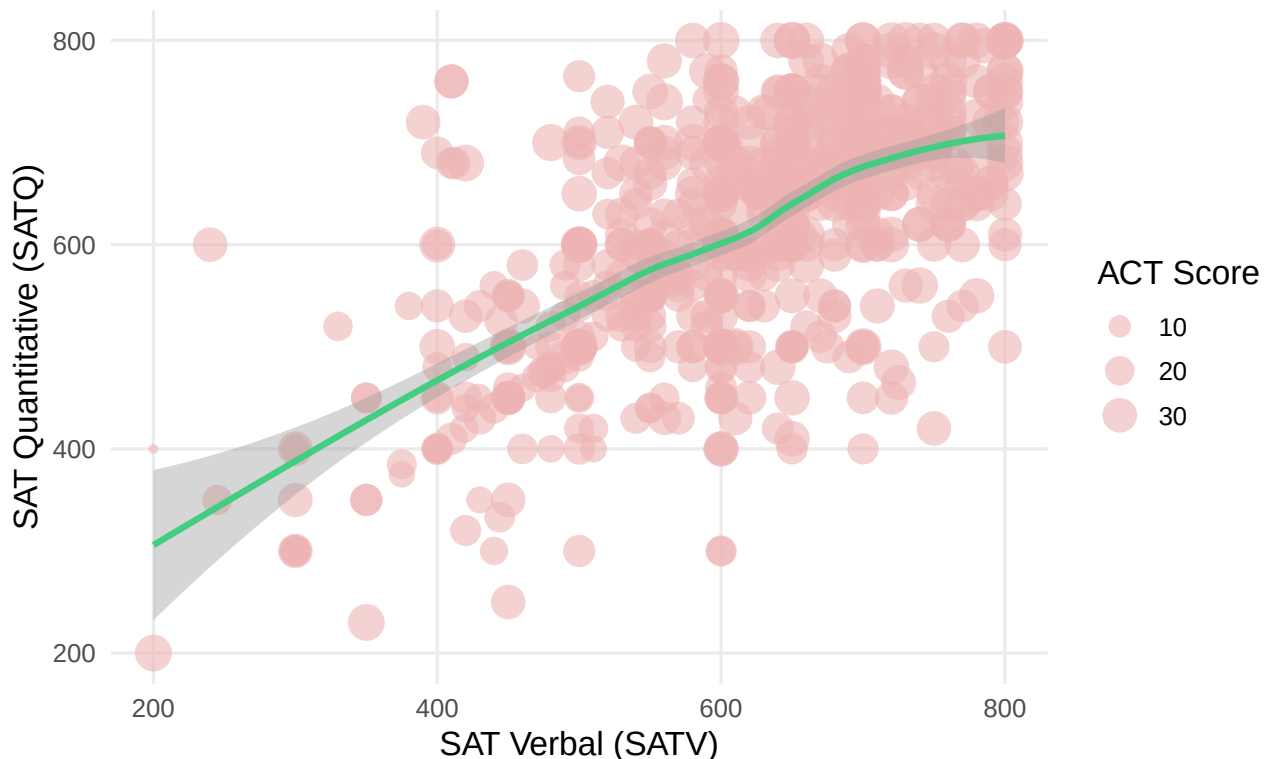
- Add a theme

```
#YOUR CODE HERE
sat_data <- psych::sat.act %>%
  # standardizes column names to lowercase (satv, satq, act) using janitor!
  clean_names()

# i used this same geom_point for my fyp haha
# visualize SATV vs SATQ with ACT as size
ggplot(sat_data, aes(x = satv, y = satq, size = act)) +
  geom_point(alpha = 0.6, color = "rosybrown2") +
  geom_smooth(
    method = "loess", se = TRUE, color = "seagreen3",
    show.legend = FALSE) + # hide from legend
  labs(
    title = "Relationship Between SAT Verbal and Quantitative Scores",
    subtitle = "Point size represents ACT composite score",
    x = "SAT Verbal (SATV)",
    y = "SAT Quantitative (SATQ)",
    size = "ACT Score"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", color = "black"),
    plot.subtitle = element_text(color = "gray30"),
    panel.grid.minor = element_blank()
  )
)
```

Relationship Between SAT Verbal and Quantitative Scores

Point size represents ACT composite score



Describe the Result: It seems like there is a positive, non-linear association between the distribution of ACT scores across SATQ and SATV scores, such that people who score high on the ACT may also score high on the SATQ and SATV. —

10) Harman74: Correlogram with Numbers

Data: psych::Harman74.cor (correlation matrix + N)

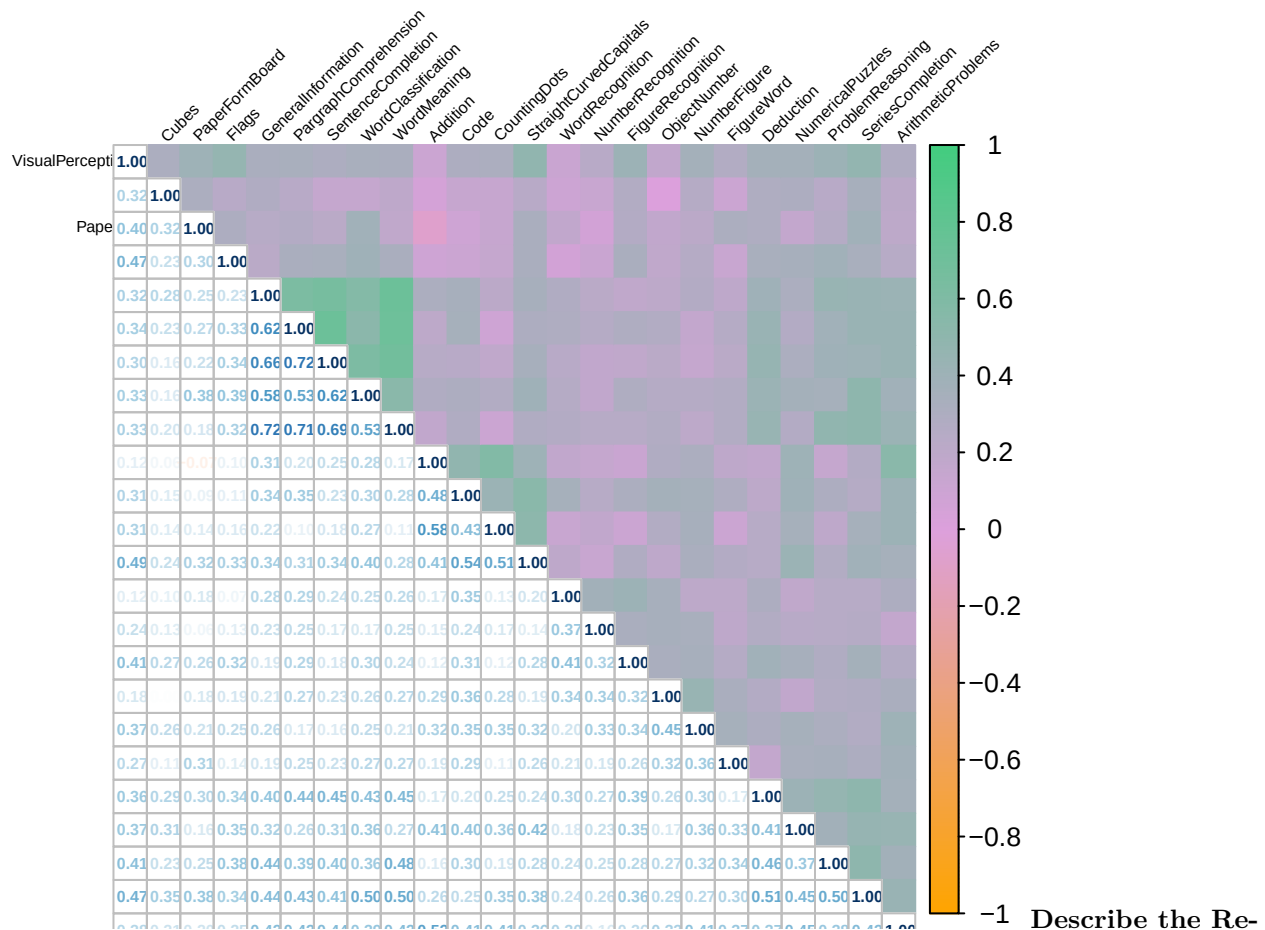
Goal: Plot a **mixed correlogram**: symbols/colors on one triangle and numbers on the other.

```
data(Harman74.cor)
harman_cor <- Harman74.cor$cov

# custom colors
corr_colors <- colorRampPalette(c("orange", "plum", "seagreen3"))(200)

# create correlogram: colors (upper), coefficients (lower)
corrplot(harman_cor,
  method = "color",
  type = "upper",
  col = corr_colors,          # custom color palette
  tl.col = "black",          # label color
  tl.srt = 45,               # rotate text labels
  diag = FALSE,
  tl.cex = .5 # no diagonal
)

# add numeric correlations on the lower triangle
corrplot(
  harman_cor,
  add = TRUE,                # overlay on existing plot
  type = "lower",            # lower half
  method = "number",         # show numeric values
  tl.pos = "n",              # suppress labels this time
  tl.cex = 1,
  number.cex = 0.5)          # adjust size of numbers
```



result: This is quite a crowded correlogram! —

Part 3: Reshaping and Comparing (msq and bfi Datasets)

The msq (Motivation State Questionnaire) dataset is in a “wide” format, with different mood scales as columns.

11) Tidying the msq Data

Data: psych::msq

Goal The msq dataset is wide. Use pivot_longer() to make it tidy.

Wrangling: Use pivot_longer() to gather all the mood scale columns (from Active to Sleepy) into two new columns: mood_scale and score.

Plotting: No plot needed. Just show the head() of your new tidy msq_tidy data frame.

```
#YOUR CODE HERE
library(psychTools)
# psych::msq doesn't exist in psych package so we need to download psychTools
msq_tidy <- psychTools::msq %>%
  pivot_longer(
    cols = active:sleepy,      # gather all mood scale columns
    names_to = "mood_scale",   # new column for mood names
```

```

    values_to = "score"      # new column for their values
  )

# first few rows
head(msq_tidy)

## # A tibble: 6 x 37
##   sluggish sociable sorry still strong surprised tense tired tranquil unhappy
##   <int>    <int> <int> <int> <int>    <int> <int> <int>    <int>    <int>
## 1      1      1    0     1     1      1     1     1      1      1
## 2      1      1    0     1     1      1     1     1      1      1
## 3      1      1    0     1     1      1     1     1      1      1
## 4      1      1    0     1     1      1     1     1      1      1
## 5      1      1    0     1     1      1     1     1      1      1
## 6      1      1    0     1     1      1     1     1      1      1
## # i 27 more variables: upset <int>, vigorous <int>, wakeful <int>,
## #   warmhearted <int>, wide.awake <int>, alone <int>, kindly <int>,
## #   scornful <int>, EA <int>, TA <int>, PA <int>, NegAff <int>,
## #   Extraversion <dbl>, Neuroticism <dbl>, Lie <dbl>, Sociability <dbl>,
## #   Impulsivity <dbl>, MSQ_Round <int>, ID <dbl>, condition <int>,
## #   MSQ_Time <dbl>, TOD <dbl>, TOD24 <dbl>, scale <fct>, exper <fct>,
## #   mood_scale <chr>, score <int>

```

Describe the Result: The tibble displays the first few rows of our new mood_scale and score columns, as well as the other columns from the msq dataset. —

12) Faceted Density of Moods

Data: Your msq_tidy data from #9

Goal: Using your msq_tidy data, create faceted density plots for the “Calm”, “Energetic”, and “Tense” mood scales.

Wrangling: Start with msq_tidy. filter() so that mood_scale is one of “Calm”, “Energetic”, or “Tense”.

Plotting: Use geom_density(aes(x = score, ...)).

- Add facet_wrap(~ mood_scale).
- Use labs() and theme_minimal(). Remove the legend.

```

#YOUR SCORE HERE
msq_subset <- msq_tidy %>%
  # filter so that we only have calm, energetic, and tense DOESN'T EXIST so
  # we use nervous instead
  filter(mood_scale %in% c("calm", "energetic", "nervous"))

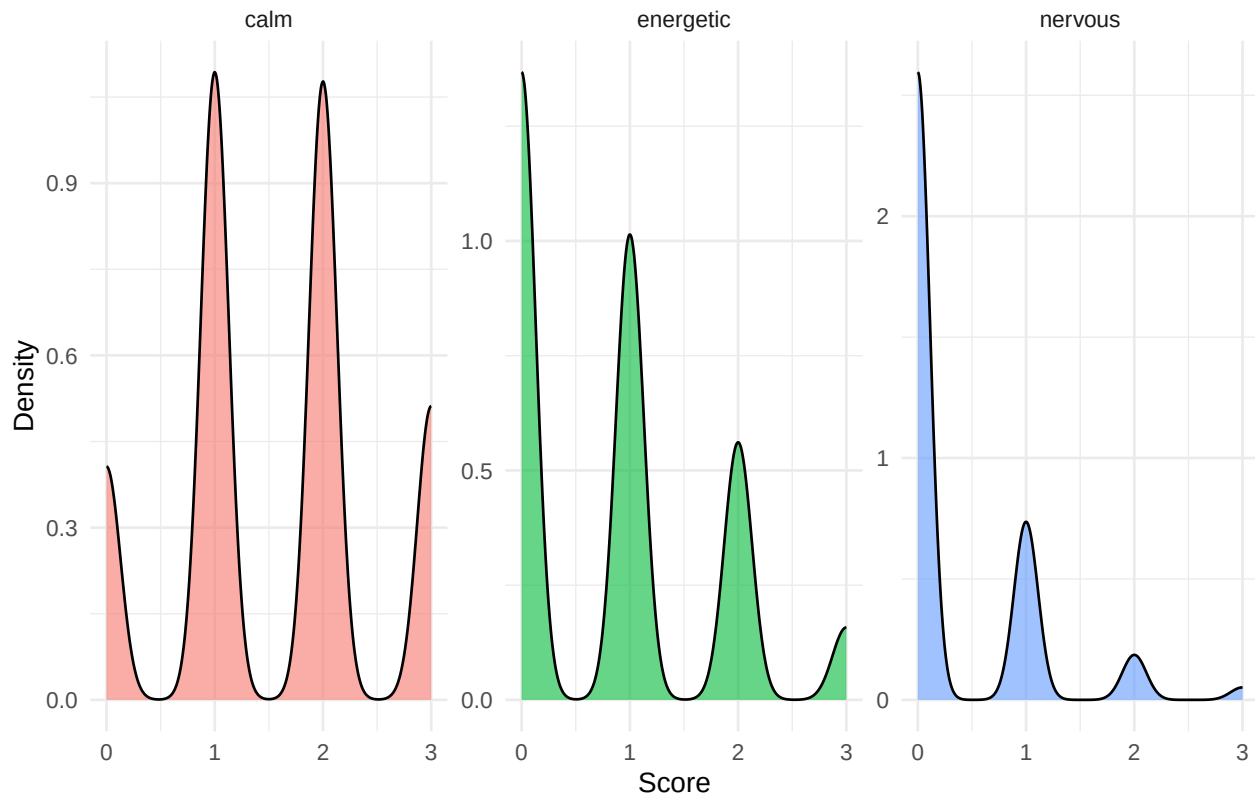
# density plots
ggplot(msq_subset, aes(x = score, fill = mood_scale)) +
  geom_density(alpha = 0.6) +
  facet_wrap(~ mood_scale, scales = "free_y") +
  labs(
    title = "Distributions of Selected Mood Scales",
    x = "Score",
    y = "Density"
  ) +

```



```
theme_minimal() +
theme(legend.position = "none")
```

Distributions of Selected Mood Scales



Describe the Result: The faceted density plots are showing how the distribution of calm, energetic, and nervous mood scores are quite different. For example, people scored 1 and 2 most for calm, while others commonly scored 1 for energetic. —

13) All BFI Traits by Gender

Data: `psych::bfi`

Goal: Create a grouped bar chart comparing the mean scores for all five BFI traits (A, C, E, N, O) grouped by gender.

Wrangling: This will be a multi-step process.

- Start with `bfi`.
- `mutate()` to create 5 new mean-score columns (`agree`, `consc`, `extra`, `neuro`, `open`).
- `select()` only the gender column and your 5 new score columns.
- `pivot_longer()` to gather the 5 score columns into `trait` and `score`.
- `group_by(gender, trait)` and `summarise(mean_score = mean(score, na.rm = TRUE))`.
- `filter()` out NA genders and `mutate(gender = factor(gender, labels = c("Male", "Female")))`.

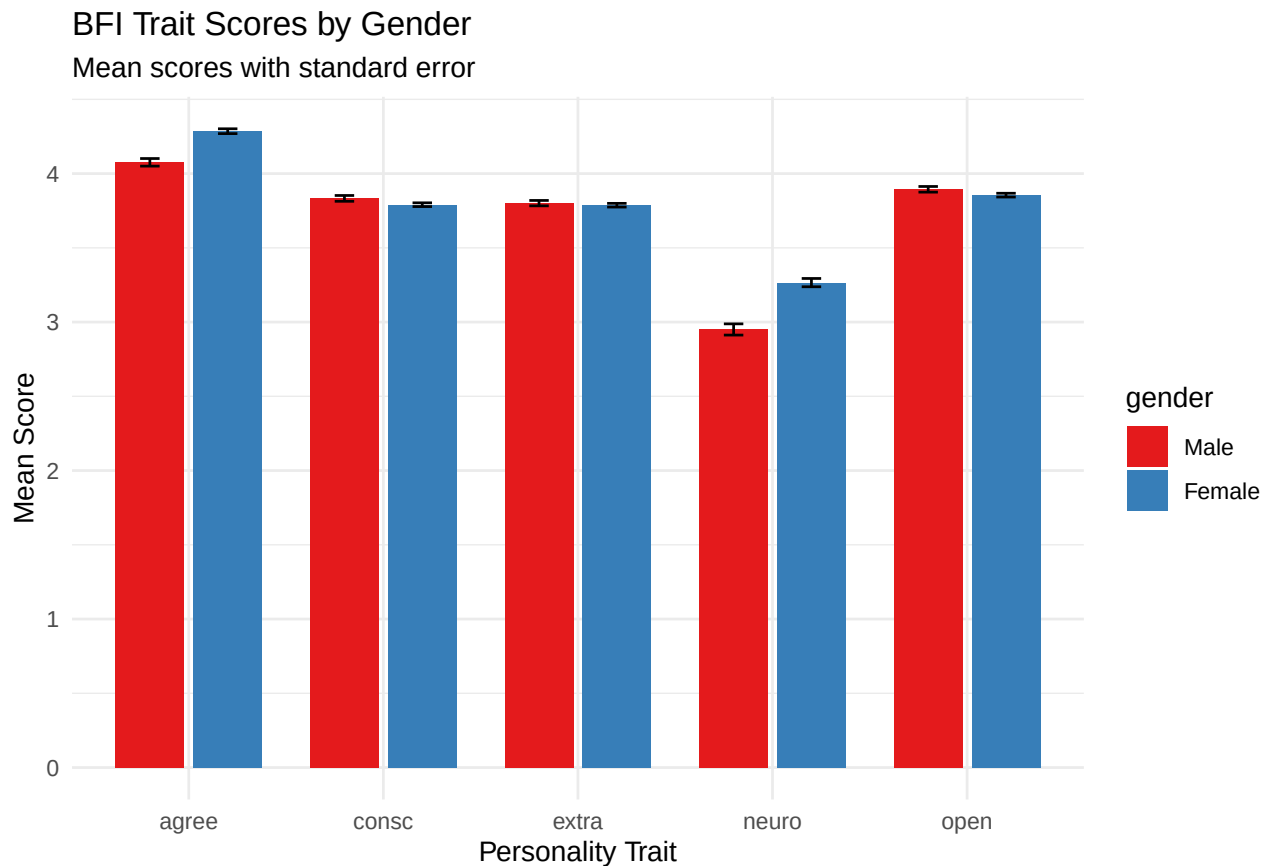
Plotting:

- `ggplot(aes(x = trait, y = mean_score, fill = gender))`.
- Use `geom_col(position = "dodge")`.

- Use `geom_errorbar()` to display your choice of measure of uncertainty.
- Use `labs()` (title, subtitle, x="Personality Trait", y="Mean Score").
- Use `scale_fill_brewer(palette = "Set1")` for a nice color palette.
- Use `theme_minimal()`.

```
#YOUR CODE HERE
bfi_tidy <- psych::bfi %>%
  # create mean scores for each trait
  mutate(
    agree = rowMeans(select(., A1:A5), na.rm = TRUE),
    consc = rowMeans(select(., C1:C5), na.rm = TRUE),
    extra = rowMeans(select(., E1:E5), na.rm = TRUE),
    neuro = rowMeans(select(., N1:N5), na.rm = TRUE),
    open = rowMeans(select(., O1:O5), na.rm = TRUE)
  ) %>%
  # keep only gender and new trait columns
  select(gender, agree, consc, extra, neuro, open) %>%
  # pivot to long format
  pivot_longer(
    cols = agree:open,
    names_to = "trait",
    values_to = "score"
  ) %>%
  # remove missing genders and convert to factor
  filter(!is.na(gender)) %>%
  mutate(gender = factor(gender, labels = c("Male", "Female"))) %>%
  # group and summarize mean and standard error
  group_by(gender, trait) %>%
  summarise(
    mean_score = mean(score, na.rm = TRUE),
    se = sd(score, na.rm = TRUE)/sqrt(n()),
    .groups = "drop"
  )

# plot
ggplot(bfi_tidy, aes(x = trait, y = mean_score, fill = gender)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean_score - se, ymax = mean_score + se),
    position = position_dodge(width = 0.8), width = 0.2) +
  labs(
    title = "BFI Trait Scores by Gender",
    subtitle = "Mean scores with standard error",
    x = "Personality Trait",
    y = "Mean Score"
  ) +
  scale_fill_brewer(palette = "Set1") + #i wanted to use pink and purple :(
  theme_minimal()
```



Describe the Result: Our plot indicates that there are slight gender differences for the mean scores on agreeableness and neuroticism. The other traits seem to have negligible gender differences among participants.

Part 4: Open Ended

14) BFI: Age Trends in Traits

Data: `psych::bfi`

Goal: For any **two traits** of your choice, examine how scores change with **age**.

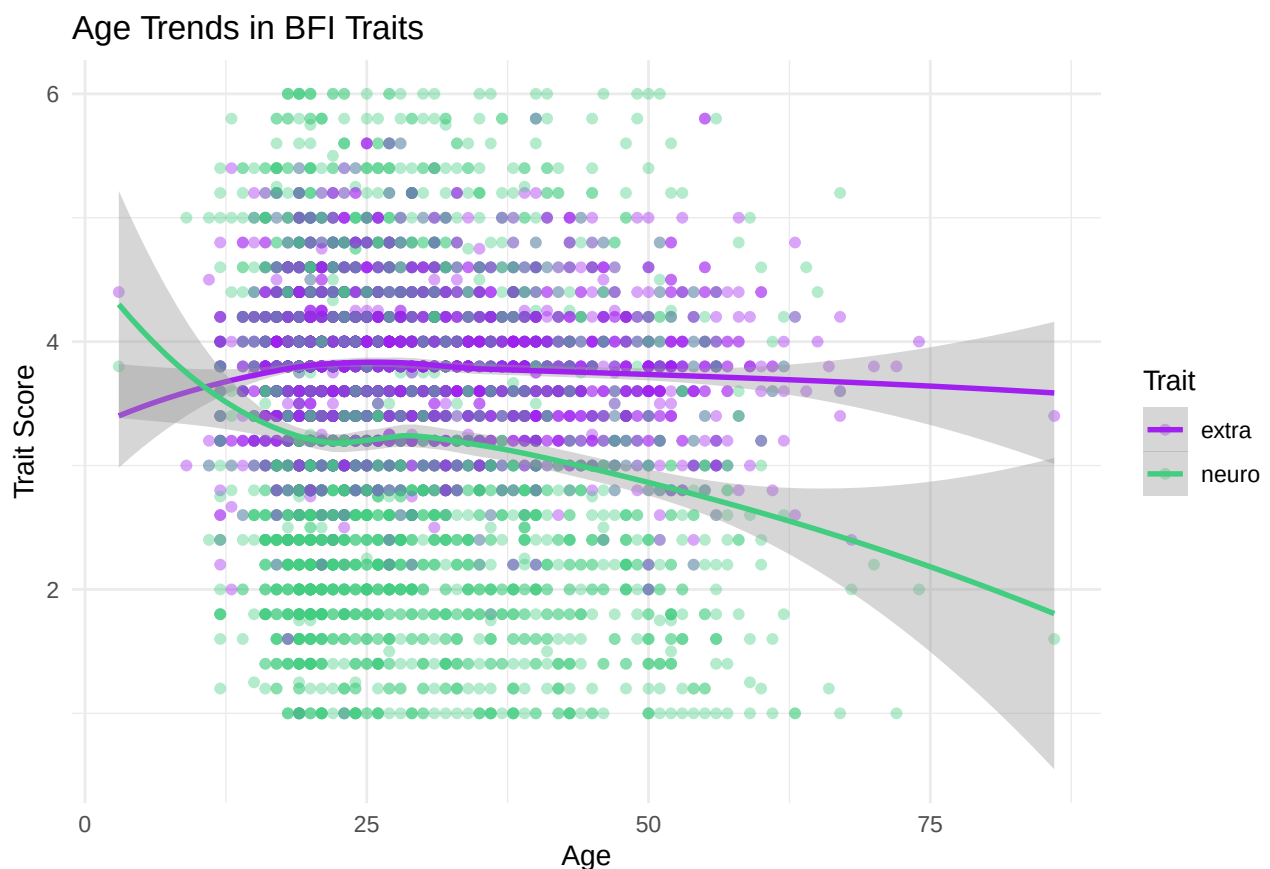
```
#YOUR CODE HERE
# let's see how extraversion and neuroticism scores change w age!
bfi_age <- psych::bfi %>%
  # compute mean scores for two traits
  mutate(
    extra = rowMeans(select(., E1:E5), na.rm = TRUE),
    neuro = rowMeans(select(., N1:N5), na.rm = TRUE)
  ) %>%
  select(age, extra, neuro) %>%
  filter(!is.na(age)) %>% # remove missing ages
  pivot_longer(
    cols = extra:neuro,
    names_to = "trait",
```

```

    values_to = "score"
  )

# plot trends
ggplot(bfi_age, aes(x = age, y = score, color = trait)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_smooth(method = "loess", se = TRUE) +
  scale_color_manual(values = c("extra" = "purple", "neuro" = "seagreen3")) +
  labs(
    title = "Age Trends in BFI Traits",
    x = "Age",
    y = "Trait Score",
    color = "Trait"
  ) +
  theme_minimal()

```



Describe the Result: It seems like extraversion doesn't change with age, but neuroticism scores seem to decrease with age, especially past 50 years old. —

15) SAT & ACT: Choose the View

Data: `psych::sat.act`

Goal: Decide on an **appropriate plot** to communicate how **SATQ** relates to **ACT**, conditioned on **SATV** (e.g., faceting SATV bins or using color gradients). Justify your choice.

```

#YOUR CODE HERE
sat_tidy <- psych::sat.act %>%

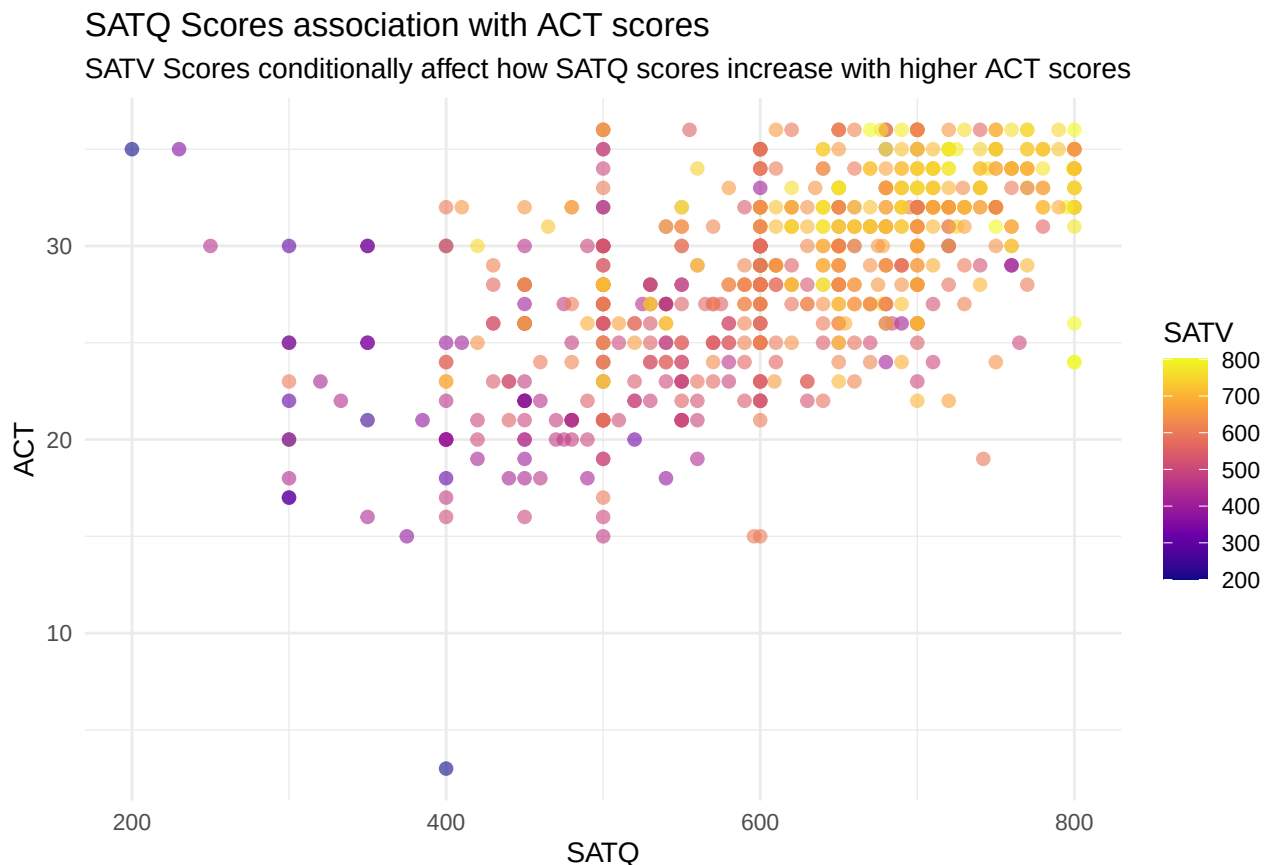
```

```

clean_names() # janitor can standardize column names: satv, satq, act

# we should use a scatter plot with a gradient
# this way we can keep all data points in one plot
# we see how SATV values influence the ACT-SATQ relationship too
ggplot(sat_tidy, aes(x = satq, y = act, color = satv)) +
  geom_point(alpha = 0.6, size = 2) +
  scale_color_viridis_c(option = "plasma") +
  labs(
    title = "SATQ Scores association with ACT scores",
    subtitle = "SATV Scores conditionally affect how SATQ scores increase with higher ACT scores",
    x = "SATQ",
    y = "ACT",
    color = "SATV"
  ) +
  theme_minimal()

```



Describe the Result: Our gradient scatter plot shows how higher SATV values also correspond to higher SATQ and ACT scores, indicating that high scores on the ACT and SATQ is also conditional on SATV. —

16) BFI: Trait Interactions by Gender

Data: psych::bfi

Goal: Investigate whether the relationship between two traits (you choose) **differs by gender**.

#YOUR CODE HERE

let's see how extraversion and neuroticism differ by gender, instead of age!

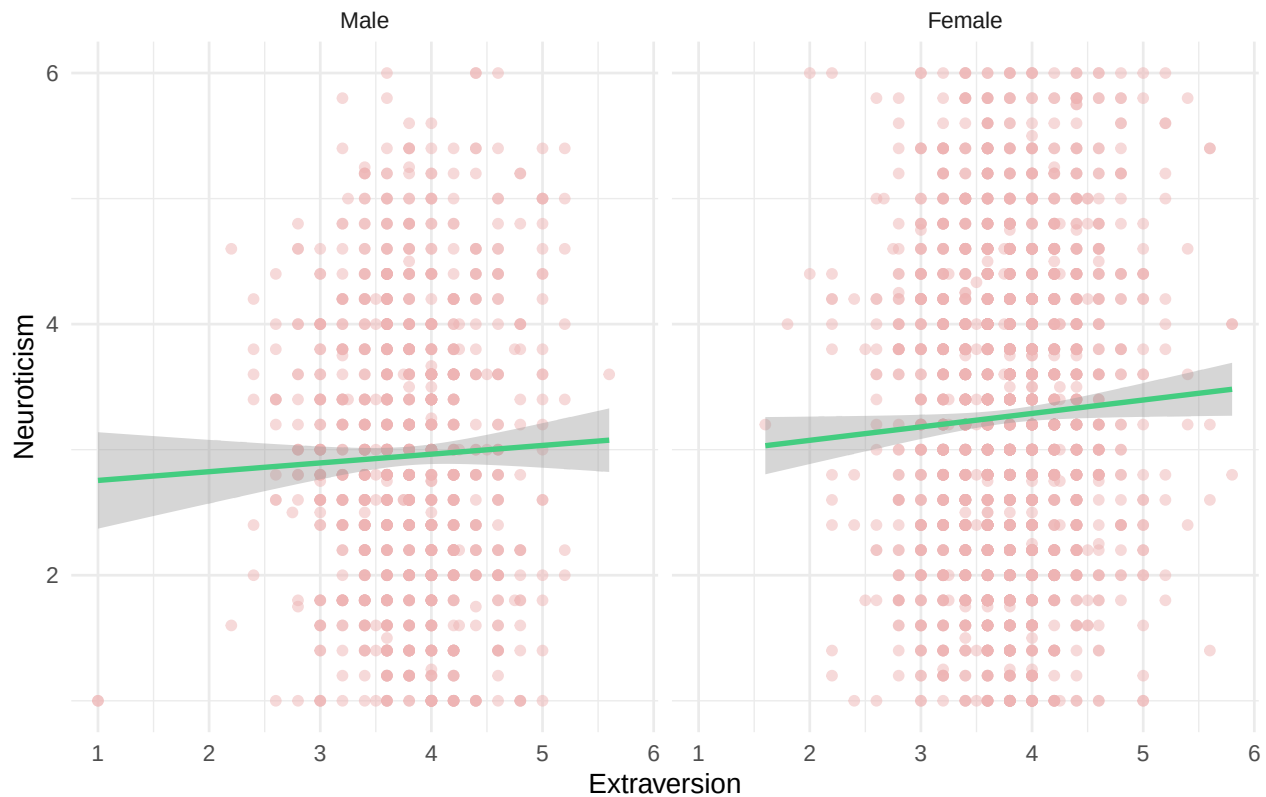
```

bfi_traits <- psych::bfi %>%
  mutate(
    extra = rowMeans(select(., E1:E5), na.rm = TRUE),
    neuro = rowMeans(select(., N1:N5), na.rm = TRUE)
  ) %>%
  select(gender, extra, neuro) %>%
  filter(!is.na(gender)) %>%
  mutate(gender = factor(gender, labels = c("Male", "Female")))

# scatter plot faceted by gender
ggplot(bfi_traits, aes(x = extra, y = neuro)) +
  geom_point(alpha = 0.5, color = "rosybrown2") +
  geom_smooth(method = "lm", se = TRUE, color = "seagreen3") +
  facet_wrap(~ gender) +
  labs(
    title = "Association between Extraversion and Neuroticism by Gender",
    x = "Extraversion",
    y = "Neuroticism"
  ) +
  theme_minimal()

```

Association between Extraversion and Neuroticism by Gender



Describe the Result: There appears to be a positive, linear association between extraversion and neuroticism scores. It does not seem like this association differs greatly between men and women.

17) “Tell a Data Story”

Use your own data. Ask a meaningful question, **tidy** the data, and produce **1–2 plots** that answer it. Write a short caption (3–5 sentences) explaining your findings and why the charts fit the question.

If you do not have any of your own data, you can pick a dataset from psych (bfi, msq, msqR, sat.act, ability.cov, Harman74.cor), but your analysis must be sufficiently different from those covered in this assignment!.

Your “story” should include:

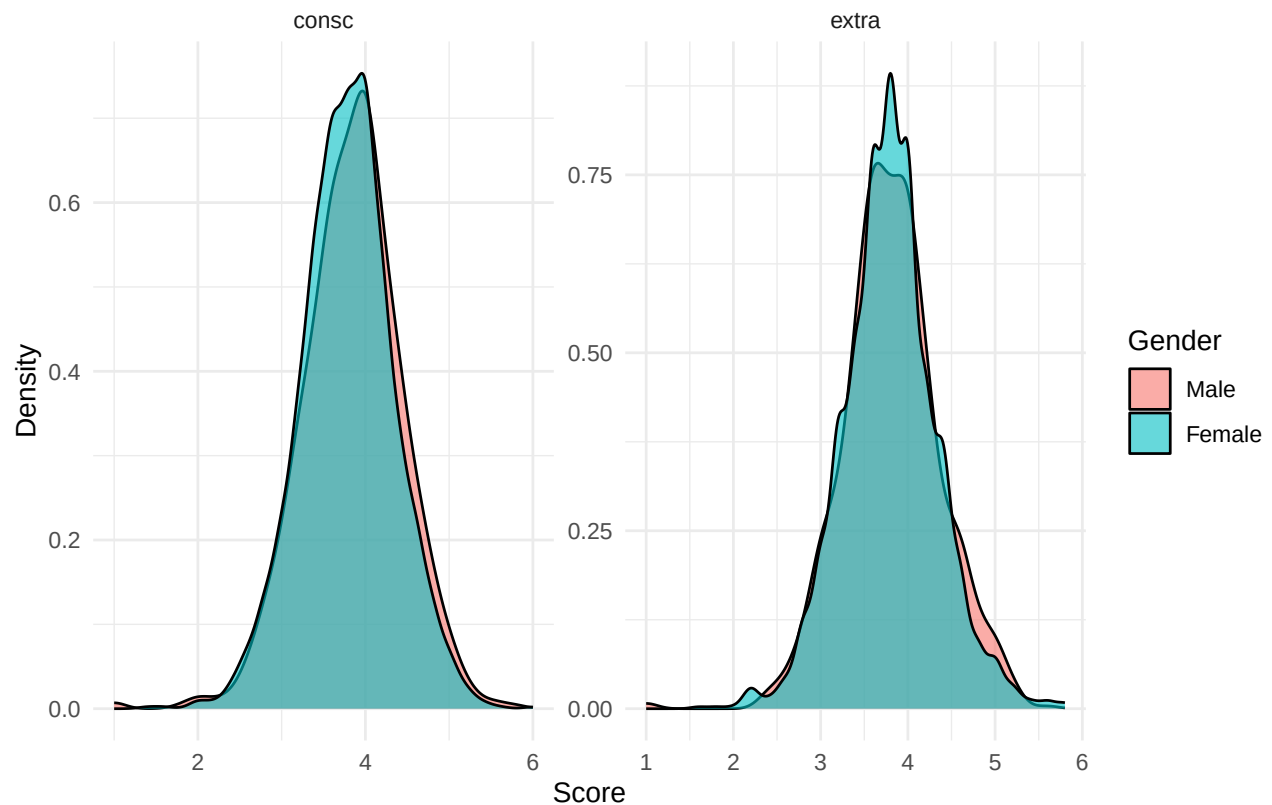
- At least two continuous DVs and two categorical IVs with 2 or more levels
- Publication quality plots of the data distributions (use `aes()` and/or faceting)
- Publication quality plots of the data summaries with estimates of uncertainty (e.g., `geom_col`)
- All appropriate labels

Kat’s Research Question: How do mean conscientiousness and extraversion scores differ by Gender and Education level?

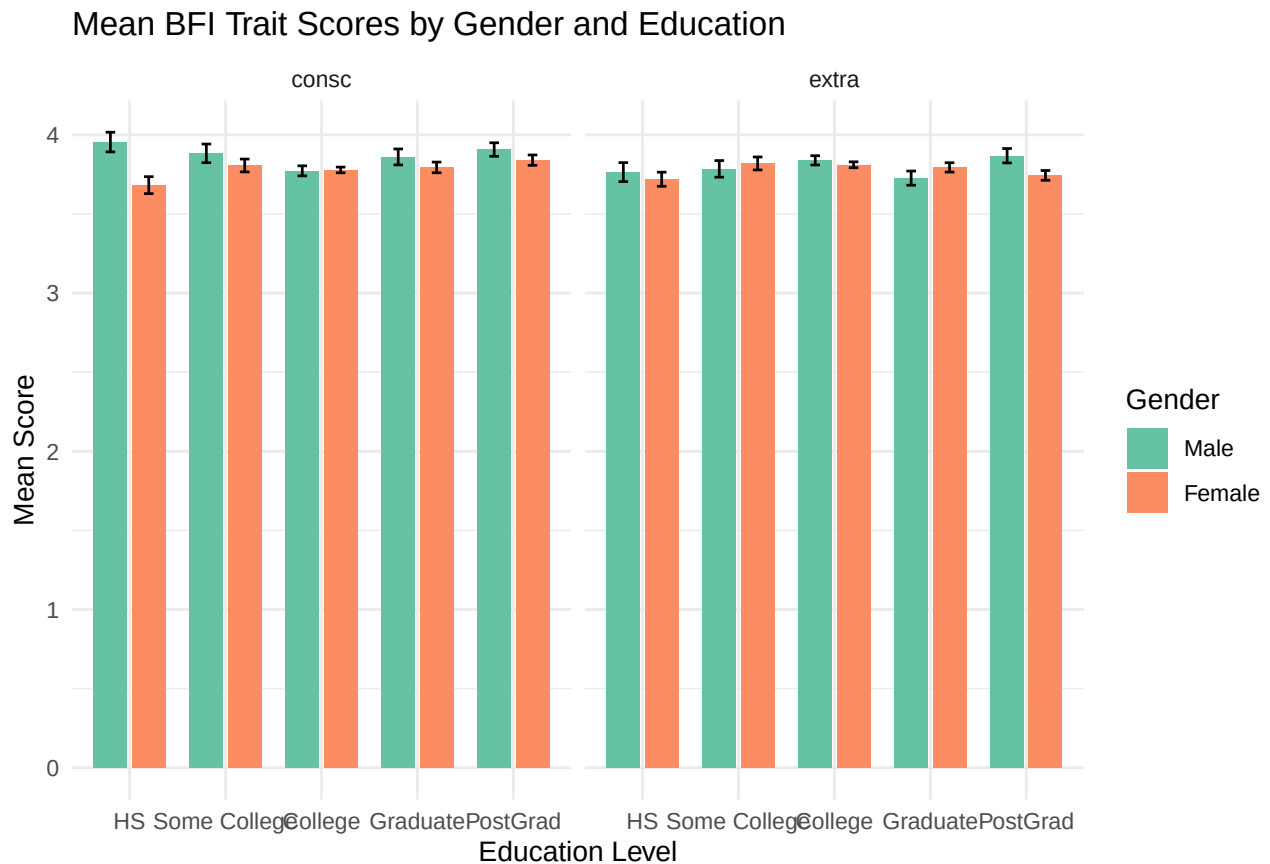
```
# TODO: choose data, tidy, visualize, and caption
# there was no way to do this w BB data unless i loaded in all of the cleaning
# codes we did too
# we will just use our BB plot codes on BFI data
bfi_research <- psych::bfi %>%
  mutate(
    gender = factor(gender, labels = c("Male", "Female")),
    consc = rowMeans(as.data.frame(select(., C1:C5)), na.rm = TRUE),
    extra = rowMeans(as.data.frame(select(., E1:E5)), na.rm = TRUE)
  ) %>%
  select(gender, education, consc, extra) %>%
  filter(!is.na(gender) & !is.na(education)) %>%
  mutate(
    education = factor(education,
                       levels = 1:5,
                       labels = c("HS", "Some College", "College", "Graduate", "PostGrad"))
  ) %>%
  pivot_longer(cols = c(consc, extra),
               names_to = "trait",
               values_to = "score")

# density faceted by trait, colored by gender
ggplot(bfi_research, aes(x = score, fill = gender)) +
  geom_density(alpha = 0.6) +
  facet_wrap(~ trait, scales = "free") +
  labs(
    title = "Distribution of BFI Traits by Gender",
    x = "Score",
    y = "Density",
    fill = "Gender"
  ) +
  theme_minimal()
```

Distribution of BFI Traits by Gender



```
# summary plot for gender and education
bfi_summary <- bfi_research %>%
  group_by(trait, gender, education) %>%
  summarise(
    mean_score = mean(score, na.rm = TRUE),
    se = sd(score, na.rm = TRUE)/sqrt(n()),
    .groups = "drop"
  )
ggplot(bfi_summary, aes(x = education, y = mean_score, fill = gender)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean_score - se, ymax = mean_score + se),
    position = position_dodge(width = 0.8), width = 0.2) +
  facet_wrap(~ trait) +
  labs(
    title = "Mean BFI Trait Scores by Gender and Education",
    x = "Education Level",
    y = "Mean Score",
    fill = "Gender"
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

Describe the Result: Our analysis examines how conscientiousness and extraversion scores vary by gender and education level. The density plots reveal that women generally score slightly higher on extraversion, while conscientiousness distributions are similar across genders. The bar plots show that conscientiousness slightly increases with education for both genders, whereas extraversion is relatively stable. Overall, it does not seem like education nor gender significantly influences mean extraversion or conscientiousness scores, but that there are subtle differences among these groups. –

About the datasets in the psych package

Here's a summary of some prominent datasets within the psych package:

bfi (Big Five Inventory):

- Content: Contains item responses from 2800 participants on 25 personality self-report items designed to measure the Big Five personality traits (Agreeableness, Conscientiousness, Extraversion, Neuroticism, Openness). Also includes demographic variables like gender, education, and age.

epi.bfi:

- Content: Data from 231 participants who completed both the Eysenck Personality Inventory (EPI - measuring Extraversion and Neuroticism) and the Big Five Inventory (bfi items).

sat.act:

- Content: Self-reported scores on the SAT Verbal (SATV), SAT Quantitative (SATQ), and ACT for 700 students. Also includes gender and education level.

How to Explore All Datasets in psych: You can see the full list of datasets available in the currently loaded version of the psych package and access their documentation directly within R:

```
# Make sure the psych package is loaded
```

```
library(psych)
```

```
# List all datasets in the psych package
```

```
data(package = "psych")
```

```
# Get detailed help documentation for a specific dataset (e.g., bfi)
```

```
?bfi
```

```
## Help on topic 'bfi' was found in the following packages:
```

```
##
```

```
##   Package           Library
```

```
##   psychTools        /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library
```

```
##   psych             /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library
```

```
##
```

```
##
```

```
## Using the first match ...
```

```
help(bfi)
```

```
## Help on topic 'bfi' was found in the following packages:
```

```
##
```

```
##   Package           Library
```

```
##   psychTools        /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library
```

```
##   psych             /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library
```

```
##
```

```
##
```

```
## Using the first match ...
```